Lab brief: does mass slow the change?

Same driving force. Different masses. Your data decides.

11 A

TASK 1 · CONTROLS CHECK

- ☐ hanging mass stays the same
- ☐ track is level
- ☐ start from rest, same spot
- ☐ release with no push
- ☐ counted cart + added + hanger
- ☐ 3 trials per mass

TASK 2 · MY DATA

TOTAL MASS (kg)	TRIAL 1 a	TRIAL 2 a	TRIAL 3 a	AVERAGE a	m × a

TASK 3 · GRAPH: AVERAGE a vs. MASS

TASK 4 · CHECK THE PRODUCT

Is m × a about the same in every row? What does that number represent?

ANALYST SIGN-OFF · CLAIM FROM MY DATA

When system mass increased, acceleration ____. Evidence: ____ vs. ____.

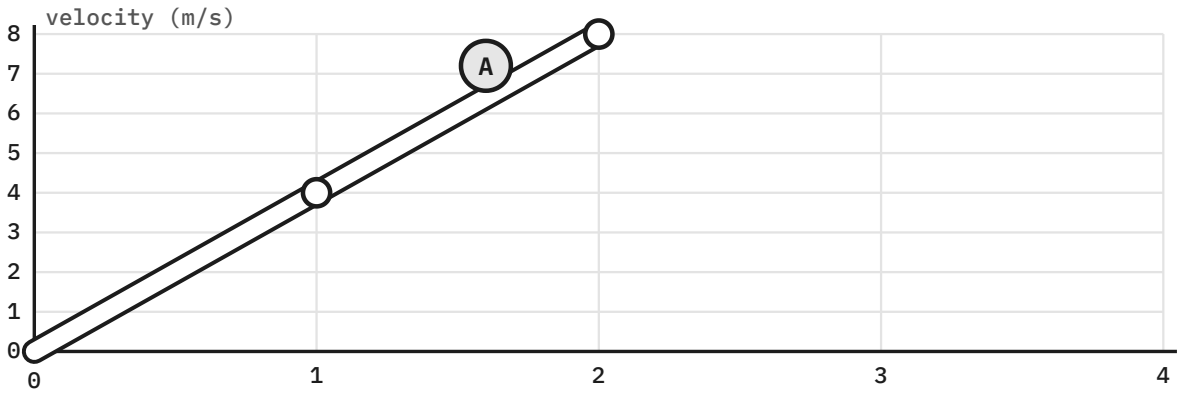
S3 1 2 3

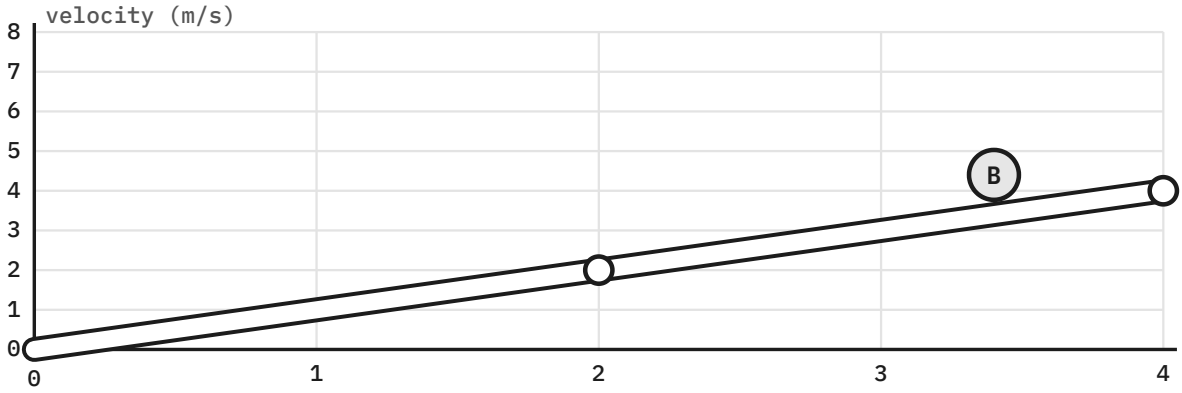
UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Read the sample lab graph

11 F

DIRECTIONS · Illustrative data (not yours): same 4 N net force, different masses.





A · 1 kg CART

slope a = ____ m/s²

B · 4 kg CART

slope a = ____ m/s²

PATTERN

mass × 4 → a × ____

EXPLAIN THE PATTERN

Use inertia: why does the heavier cart gain velocity more slowly?

S3 1 2 3

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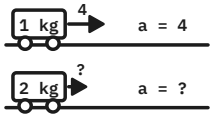
UNIT 1 · DAY 11 · S3 · SEI · PARTNER TALK

Talk it through: the mass lab

Language goal: I can **make a claim** from data.

DIRECTIONS · Partner A and B read their lines aloud. Listen, then write. Switch each round.

ROUND 1 · 1 kg → 4 m/s², 2 kg → ? (SAME FORCE)



A ASKS "What happens to a when the mass doubles?"

B SAYS "The mass goes from ____ to ____ kg, so a goes from 4 to ____ m/s²."

A CHECKS "So you're saying more mass gives ____ acceleration."

ROUND 2 · SWITCH · WHAT STAYED THE SAME?



stays the same

B ASKS "Why did we keep the hanging mass the same?"

A SAYS "To keep the ____ force the same. Then only the ____ changes."

B ADDS "I want to add that the hanger is part of the ____ mass."

ROUND 3 · BOTH · CHECK m × a

1	4
2	2
4	1

×2 ÷2

A "1 × 4 = ____ and 2 × 2 = ____."

B "The product stays ____, so it equals the ____ force."

LISTENING LOG · MY PARTNER SAID…

NOW I WRITE IT MYSELF

When mass increases, acceleration…

Stems: I agree because… · Can you say that again? · What does ___ mean? · I want to add…

S3 (1) (2) (3)

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

UNIT 1 · DAY 11 · S3 · SEI · SENTENCE FRAMES

Say it, then write it: lab claim

Language goal: I can **make a claim** from data.



mass
masa · massa
amount of matter (kg)



control
control · controle
what stays the same



trial
prueba · ensaio
one run of the test



pattern
patrón · padrão
what repeats in data

English · Spanish · Portuguese. My home language: _____

1 · READ AND FILL · WORD BANK: mass · control · trial · decreased · evidence

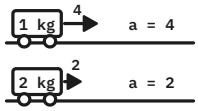
In each ____, we changed only the ____. The hanging mass was the ____. When mass went up, acceleration ____. My ____ is two measured values.

2 · SAY IT TO A PARTNER, THEN WRITE IT



When the mass was ____ kg, the acceleration was ____ m/s².
When mass increased, acceleration ____ because ____ .

3 · SOLVE WITH THE STEPS



Sample: 1 kg → 4 m/s². 2 kg → 2 m/s².

STEP 1 · COMPARE	STEP 2 · PRODUCT	STEP 3 · SAY IT
mass × ____ · a × ____	1 × 4 = ____ · 2 × 2 = ____	"More mass, ____ acceleration."

4 · EXPLAIN IT · CHOOSE A FRAME

- ☐ My claim is ____ because my data shows ____ .
- ☐ One control was ____ so the test was fair.
- ☐ A trial that did not fit was ____ because ____ .

I said it ☐ · I wrote it ☐

S3 1 2 3

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UNIT 1 · DAY 11 · S3 · CHOICE BOARD

Mass lab tic-tac-toe

11 D

DIRECTIONS · Complete three in a row. Shade each square you finish.

CALCULATE v rises 0 → 4 m/s in 2 s. Find a.	**DRAW** Sketch two v–t lines: light cart and heavy cart, same force.	**EXPLAIN** Why must the hanging mass stay the same today?
REAL WORLD Why is a loaded truck slower to speed up than an empty one?	**CHALLENGE ★** m × a ≈ 4 N in every row. Predict a for a 5 kg system.	**CALCULATE** 1 kg: 4 m/s². Predict a at 2 kg with the same force.
EXPLAIN What counts as "system mass" in the hanger setup?	**DRAW** Plot a vs. m for 1, 2, 4 kg at 4, 2, 1 m/s².	**REAL WORLD** 2026-XJ has a huge mass. What does that mean for a push?

Three in a row: _____ · The square that taught me most: _____

S3 1 2 3

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Settle the lab argument

11 I

DIRECTIONS · Shade the bubble of the person who is right. Use lab evidence.

ROUND 1

ANA
"We added mass to the cart AND the hanger, so we tested mass fairly."

BEN
"Two things changed. Keep the hanging mass fixed."

Who's right? _____ · Evidence: _____

Fixed claim: _____

ROUND 2

KAI
"The cart is the only mass that accelerates."

LU
"The hanger accelerates too — count it in the system mass."

Who's right? _____ · Evidence: _____

Fixed claim: _____

ROUND 3

MAYA
"At 1 kg we got 4 m/s², at 2 kg 2 m/s². Mass doesn't matter."

RAJ
"Doubling mass halved a. Mass matters a lot."

Who's right? _____ · Evidence: _____

Fixed claim: _____

YOUR TURN · WRITE A TRAP

Write a claim about today's idea that sounds right but is wrong. Trade and settle it.

S3 1 2 3

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